

3.4–3.5 Independence, Basis, Dimension, and Subspaces

Concise exam revision notes for independence, span, bases, and subspace dimensions.

Based on handwritten MIT 18.06 revision notes.

1 Core Idea

Core idea. Independence, span, and basis are all controlled by the equation $Ax = 0$, where the columns of A are the vectors being tested.

For vectors $v_1, \dots, v_n$, put them into a matrix

$$A = [v_1 \quad v_2 \quad \cdots \quad v_n].$$

Then the dependence test is

$$x_1v_1 + x_2v_2 + \cdots + x_nv_n = 0 \quad \Longleftrightarrow \quad Ax = 0.$$

The vectors are independent exactly when the only solution is $x = 0$.

2 Independence and Dependence

Situation	Meaning	Exam consequence
Independent columns	$Ax = 0$ has only $x = 0$.	No free variables.
Dependent columns	$Ax = 0$ has a nonzero solution.	At least one free variable.
More columns than rows	$n > m$.	Columns must be dependent.

Exam trigger. To test whether vectors are independent, solve $Ax = 0$, not $Ax = b$.

If $Ax = 0$ has a nonzero solution, that solution gives an explicit linear relation among the columns. For example,

$$x_1v_1 + x_2v_2 + x_3v_3 = 0$$

with some nonzero coefficient means one vector is forced by the others.

3 Span

The span is the set of all linear combinations of the given vectors:

$$\text{span}\{v_1, \dots, v_k\} = \{c_1v_1 + \cdots + c_kv_k\}.$$

It answers the question: which vectors can be produced?

Vectors in $\mathbb{R}^2$	Span
$(1, 0)^T, (0, 1)^T$	All of $\mathbb{R}^2$.
$(1, 0)^T, (-1, 0)^T$	Only the x -axis.

Common mistake. Having two vectors in $\mathbb{R}^2$ does not automatically mean they span $\mathbb{R}^2$. They must point in independent directions.

4 Basis

A basis is a set of vectors that is both independent and spanning.

Property	What it prevents	What it guarantees
Independent	No repeated direction.	No vector is unnecessary.
Spanning	No missing direction.	Every vector in the space is reachable.
Basis	Both at once.	Coordinates are unique.

Exam memory. A basis is the smallest spanning set and the largest independent set.

5 Row Space and Column Space

For an $m \times n$ matrix A , the column vectors have length m , while row vectors have length n .

Space	Built from	Lives in	Basis rule
Column space $C(A)$	Columns of A .	$\mathbb{R}^m$	Use pivot columns of the original A .
Row space $C(A^T)$	Rows of A .	$\mathbb{R}^n$	Use nonzero rows of echelon form R .

Example:

$$A = \begin{bmatrix} 2 & 4 \\ 3 & 6 \end{bmatrix}, \quad R = \begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix}.$$

The row space can be represented from R :

$$C(A^T) = \text{span}\{(1, 2)\}.$$

The column space must use the pivot column from the original matrix:

$$C(A) = \text{span} \left\{ \begin{bmatrix} 2 \\ 3 \end{bmatrix} \right\}.$$

Common mistake. Do not use the pivot columns of R as a basis for $C(A)$. Row operations change the columns, but they preserve the row space.

6 Dimension and Rank

The dimension of a subspace is the number of vectors in a basis for that subspace.

Quantity	Value	Meaning
$\dim C(A)$	r	Number of pivot columns.
$\dim C(A^T)$	r	Number of nonzero rows in R .
$\dim N(A)$	$n - r$	Number of free variables.
$\dim N(A^T)$	$m - r$	Number of missing pivots among rows.

The row space and column space have the same dimension:

$$\dim C(A) = \dim C(A^T) = r.$$

They usually live in different spaces, so they are not the same subspace.

7 Four Fundamental Subspaces

For $A \in \mathbb{R}^{m \times n}$, the four subspaces are:

Subspace	Lives in	Dimension	How to find a basis
Column space $C(A)$	$\mathbb{R}^m$	r	Pivot columns of A .
Row space $C(A^T)$	$\mathbb{R}^n$	r	Nonzero rows of R .
Nullspace $N(A)$	$\mathbb{R}^n$	$n - r$	Special solutions of $Ax = 0$.
Left nullspace $N(A^T)$	$\mathbb{R}^m$	$m - r$	Special solutions of $A^T y = 0$.

Dimension check. The dimensions split by ambient space: $r + (n - r) = n$ in $\mathbb{R}^n$, and $r + (m - r) = m$ in $\mathbb{R}^m$.

8 Rank-One Reminder

A rank-one matrix is one column times one row:

$$A = uv^T.$$

For example,

$$\begin{bmatrix} a \\ b \\ c \end{bmatrix} \begin{bmatrix} 2 & 3 & 7 & 8 \end{bmatrix}$$

has rank 1 if both factors are nonzero.

Exam use. Rank-one matrices are useful because all columns are multiples of one vector and all rows are multiples of one row.

9 Common Mistakes

Mistake	Correct exam habit
Testing independence with $Ax = b$.	Test with $Ax = 0$.
Forgetting that $n > m$ forces dependence.	More vectors than dimension means at least one free variable.
Using columns of R for $C(A)$.	Use pivot columns of original A .
Saying row space and column space are the same.	They have the same dimension r , but may live in different spaces.
Counting all columns as a basis.	Count pivot columns only.

10 Final Exam Checklist

Checklist.

1. Write the size: A is $m \times n$.
2. Put the vectors as columns if testing independence or span.
3. Solve $Ax = 0$ to test independence.
4. Find the rank r from pivots.
5. Use pivot columns of A for a basis of $C(A)$.
6. Use nonzero rows of R for a basis of $C(A^T)$.
7. Use special solutions for a basis of $N(A)$.
8. Check dimensions: r , r , $n - r$, and $m - r$.
9. State whether the vectors are independent, spanning, a basis, or dependent.